

RAJ PRATAP SINGH SENGAR

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PROFESSIONAL SUMMARY

Results-driven Computer Science undergraduate (B.Tech, 2024–2028, CGPA: 9.10/10) with practical experience in building intelligent, production-ready software. Specializes in end-to-end Machine Learning pipelines, Computer Vision, and AI-driven systems. Proven ability to independently architect and deliver full-stack AI projects using Python, TensorFlow, PyTorch, FastAPI, and Docker. Seeking to contribute technical expertise and a problem-solving mindset to a challenging software engineering or AI/ML role.

EDUCATION

B.Tech – Computer Science and Engineering (CGPA: 9.10 / 10.00) Amity University Madhya Pradesh Gwalior, MP	2024 – 2028
Higher Secondary Certificate (Class XII) – PCM 87% MP Board of Secondary Education (MPBSE) Physics, Chemistry, Mathematics	2021 – 2022
High School Certificate (Class X) 83% MP Board of Secondary Education (MPBSE) Gwalior, MP	2019 – 2020

PROJECTS

Network Phishing Detection System

Tech Stack: Python, FastAPI, Scikit-learn, MLflow, MongoDB, Docker | [GitHub](#)

- Engineered a production-grade, end-to-end ML pipeline to classify phishing websites using 30+ URL and domain-based security features.
- Designed modular Python components for data ingestion, validation, transformation, feature engineering, and automated model training.
- Built and deployed a real-time prediction REST API using FastAPI, enabling live phishing URL classification.
- Integrated MLflow for experiment tracking, model versioning, and performance monitoring across training runs.
- Containerized the full application stack using Docker and used MongoDB as the primary data storage layer.

Sanjeevani AI – AI-Powered Elderly Fall & Emergency Detection System

Tech Stack: Python, OpenCV, YOLOv8, MediaPipe, NumPy, SciPy, SpeechRecognition, SoundDevice | [GitHub](#)

- Developed a real-time elderly safety monitoring system using YOLOv8, MediaPipe, and OpenCV for fall and emergency detection.
- Implemented person detection, pose estimation, and posture classification for Standing, Sitting, Lying, and Lying Moving states.
- Built rule-based fall detection using body angle, movement, angular velocity, posture transition, and temporal buffering.
- Added lying-still danger detection to identify unconscious or motionless persons after a fall.
- Integrated Sanjeevani voice assistant to trigger emergency alerts using voice commands such as 'Sanjeevani need help'.
- Implemented automatic emergency video recording and timestamp-based alert logging for fall, danger, recovery, and voice events.

- Created manual bed-zone detection using polygon ROI to identify whether the person is ON BED or OFF BED.

TECHNICAL SKILLS

Programming Languages: C++, Python

Machine Learning: Data Preprocessing & Feature Engineering, Model Training

Deep Learning & AI: Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN)

Computer Vision: OpenCV, MediaPipe, yolov8

Backend Development: FastAPI, Flask

Databases: MongoDB, MySQL

DevOps & Tools: Git, GitHub, Docker, MLflow

Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow

Core CS Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Operating Systems

CERTIFICATIONS

- Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp — Udemy | Instructor: Krish Naik, KRISHAI Technologies Pvt. Ltd. | 99 hours | March 2026
- Data Structures and Algorithms (DSA) — Apna College Alpha Batch | Comprehensive DSA course covering Arrays, Linked Lists, Trees, Graphs, Dynamic Programming in C++

ACHIEVEMENTS & ADDITIONAL INFORMATION

- Maintaining a CGPA of 9.10/10 in B.Tech CSE at Amity University Madhya Pradesh (2024–2028).
- Solved 140+ problems on LeetCode across Arrays, Strings, Dynamic Programming, Trees, and Graphs — strengthening algorithmic problem-solving in C++ and Python.
- Developed two end-to-end, production-ready ML/AI projects demonstrating full-stack data science and software engineering capability.
- Proficient in MLOps practices (MLflow) for reproducible and trackable machine learning workflows.

Declaration

I hereby declare that all information provided above is accurate and true to the best of my knowledge.

Place: Gwalior

Date: 03/06/2026

Raj Pratap Singh Sengar